UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2024/2025 SESSION
MTH 415: MATHEMATICAL METHODS IV TIME: 2 HOURS

INSTRUCTIONS: ATTEMPT 4 QUESTIONS ONLY

1. Discuss the solution of the Euler's differential equation for each of the following cases:
   a) $F = F(x, y)$
   b) $F = F(x, y')$
   c) $F = F(y, y')$

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2. Find the extremals of the following functions
   a) $F = y\sqrt{1 + y'^2}$
   b) $F = x\sqrt{1 + y'^2}$

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3. Prove that the necessary and sufficient condition that the integral

$$I(\Phi) = \int_{x_0}^{x_1} F(x, \Phi, \Phi')dx$$

Be stationary at $\Phi = u$ is that $\Phi = u$ shall be an admissible function which satisfies the Euler's differential equation

$$L(u) = F_u - \frac{d}{dx}F_{u'} = 0$$

(Any Lemma used must be clearly stated and proved).
Deduce that the necessary condition implies

$$L(u) = F_u - F_{u'u'}u'' - F_{uu'}u' - F_{xu'}.$$

4. In a vertical xy-plane, a point $A(x_0, y_0)$ is to be joined to a point $B(x_1, y_1)$, such that $x_1 > x_0$, $y_1 > y_0$, by a smooth curve $y = u(x)$ in such a way that the time taken by a particle sliding frictionless from A to B along the curve under gravity (acting in the direction of the positive y-axis) is as short as possible.
   a) What problem is described above?
   b) Show that the time taken in the fall of the particle is given by

$$T = \frac{1}{\sqrt{2g}}\int_{x_0}^{x_1} \frac{\sqrt{1 + y'^2}}{\sqrt{y - y_0}}dx$$

Dropping the unimportant factor $\frac{1}{\sqrt{2g}}$, and taking $y_0 = 0$ in the above, find among all continuously differentiable functions $y = \Phi(x), y \geq 0$, for which $\Phi(x_0) = 0, \Phi(x_1) = y_1$, the one for which the integral

$$I(\Phi) = \int_{x_0}^{x_1} \sqrt{\frac{1 + y'^2}{y}}dx$$

has the least possible value.

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5. Given a function $F(x, \Phi, \Phi')$ of three arguments, which in the region of the arguments considered is continuous and has continuous derivatives of the first and second orders. Replacing, in this function F, $y = \Phi(x)$ and Φ' by the derivative $y = \Phi'^{(x)}$, F becomes a function of x, and an integral of the form

$$I(\Phi) = \int_{x_0}^{x_1} F(x, y, y')dx$$

yields a definite number which depends on the behaviour of the function $y = \Phi(x)$.
   a) State the fundamental theorem of the calculus of variations.
   b) State Fermat's principle of least time. Hence show that, if the velocity of light is a given function $v(x, y, y')$ of the point (x, y) and the y', then the optical problem is equivalent to the general problem stated above, if we relate the functions F and v to one another by putting $F = \frac{\sqrt{1+y'^2}}{v}$

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6. Obtain the necessary conditions that a function $u = \Phi(x)$ may give an extreme value of the integral $I(\Phi) = \int_{x_0}^{x_1} F(x, y, y')dx$. Write down the first variation of the integral.